

Empirical science has two major objectives: to describe particular phenomena in the world of our experience and to establish general principles by which they can be explained and predicted. The explanatory and predictive principles of a scientific discipline ... characterize general patterns and regularities to which ... individual phenomena conform and by virtue of which their occurrence can be systematically anticipated. (Hempel, 1952, p. 653)

Background and motivations: “New Experimentalism” (Cartwright, Franklin, Galison, Hacking, ...) as the immediate context for B & W. Two recurring themes:

- Reject foundationalist pictures that grant direct sense experience a privileged role; reorient the empiricist position to capture the contribution of sense experience to scientific knowledge without that role.
- “Experiments and data analysis have lives of their own”: shift away from regarding experimentation as fundamentally about “checking predictions,” toward an account of the variety of activities that constitute scientific practice (instrumental design, exploratory experiments, statistical methods, assessments of reliability ...). Particularly salient for fields that are “data rich and theory poor.”

Secondary motivation (Woodward 2011): “heterodox” view of statistical explanation: orthodox QM explains the probabilities of measurement outcomes but not the individual outcomes themselves — yet aggregated outcomes can serve as evidence for the probability claims.

B & W’s Thesis

Our general thesis, then, is that we need to distinguish what theories explain (phenomena or facts about phenomena) from what is uncontroversially observable (data). Traditional accounts of the role of observation in science blur this distinction and, because of this, neglect or misdescribe the details of the procedures by which scientists move from claims about data to claims about phenomena. In doing so, such accounts also overlook a number of considerations which bear on the reliability of scientific knowledge. (B & W, p. 314.)

An Astronomical Illustration

Phenomenon 1 (Newton, *Principia*, 1687): “The circumjovial planets, by radii drawn to the centre of Jupiter, describe areas proportional to the times, and their periodic times ... are as the $3/2$ powers of their distances from that centre.” (Modernly: the moons of Jupiter satisfy Kepler’s laws.)

Three contrasting questions the case helps to keep separate:

- What explains *what Galileo actually saw* on a particular evening in 1610? How relevant is this to assessing the causes of the motion of Jupiter’s moons?
- What can we infer from observations like Galileo’s about the motions of the Medicean satellites, and what is required to establish the reliability of that inference? (Reliability of telescopes; understanding of optics; ability to distinguish satellites over time; calculation of orbital periods and distances; ...)
- What explains the phenomenon, once established — that Jupiter’s moons obey Kepler’s laws?

The Distinction

- *Data*: public records produced by particular acts of measurement and experiment (Eddington's photographic plates; bubble chamber photographs; thermometer readings on particular occasions). Reflect the influence of many causal factors — the target phenomenon among others — including factors idiosyncratic to the apparatus and its local environment.
- *Phenomena*: features of the world that “could in principle recur under different contexts or conditions.” Stable, repeatable, and (typically) the appropriate target of systematic explanation. Examples: the melting point of lead; the gravitational deflection of starlight; the existence of weak neutral currents; the behavioral consequences of frontal lesions.

Two Senses of Explanation

- *Systematic explanation*: detailed, principled derivation of an outcome from underlying laws and principles, with quantitative content (e.g., GR derives the magnitude of starlight deflection, not merely that gravity bends light). Theories like classical mechanics, GR, the electroweak theory aim at this. Facts about phenomena are natural targets of such explanation in a way that facts about data are not.
- *Singular causal explanation*: some causal factor is identified as contributing to the production of an outcome, without a quantitative account of how the outcome depends on this factor or what others contribute. In well-functioning measurements, the phenomenon *will* figure as one causal factor among many in the production of the data — B & W are not denying this. What they deny is that systematic explanations of individual data points are typically available or required.

Reliability vs. “Explaining the Data” (§ VI)

Central conceptual move: assessing the reliability of an inference from data to phenomena is a different matter from systematically explaining the data, and the two have been conflated in much of the philosophical literature. Several aspects of reliability assessment that do *not* require systematic explanation of data:

- *Control of confounding factors / backgrounds*. Background effects in particle physics (e.g., neutral current detection): successful methods of treating backgrounds did not embed individual data points into a theoretical model. Arguments turned on detailed features of the apparatus, simulation, and experimental design.
- *Empirical investigation of instrumental reliability*. Hacking on microscopes: reliability was established before the wave-optical theory of imaging (in which diffraction plays an unexpected role) was available.
- *Data reduction and statistical analysis*. Inference to a target quantity from a scatter of measurements requires assumptions about the error distribution, not derivations of the individual readings.

A virtue of these procedures is their relative *robustness* to changes in high-level theory: reassessments at the level of fundamental theory rarely overturn well-grounded judgments of instrumental or statistical reliability (B & W, p. 334).

Anticipated Objections and Replies

O1 *Isn't this just theory-ladenness of observation revisited?* (§§IX–XII)

The data/phenomena distinction is largely *orthogonal* to the standard theory-ladenness debate. Data in B & W's sense are typically uncontroversially observable — a thermometer reading on a particular occasion is observable on anyone's account of observation. The interesting question is not whether perception is conceptually pure but what considerations make the inference *from* such observable records *to* claims about phenomena reliable. Sections IX–XII work through several leading accounts of observation — the “new look” theories of perception, Kuhn's view of observation, and Shapere's information-theoretic proposal — and argue that none of them captures either the data/phenomena distinction or the reliability question.

O2 *Theories **do** predict and explain data, at least with suitable auxiliaries.* (§ VI; Woodward 2011, § 1)

The “in principle” derivability of d_i from theory plus suitable auxiliaries is trivial: one can always derive d_i from “if P obtains in circumstance C , then d_i ” together with “ P obtains in C .” What matters is whether scientists actually exhibit, or need to exhibit, such derivations in order for data to provide evidence about phenomena. They typically do not, and they need not: reliability is established by quite different routes (control of confounders, calibration of instruments, statistical analysis). The salient epistemic direction is upward — from data plus background assumptions to the phenomenon — not downward from theory to particular data points. Woodward (2011, § 1) develops this point in his reply to Votsis.

O3 *There is no principled distinction between “data” and “phenomena”.* (§§III–IV)

B & W concede that “phenomena” may not pick out a single ontological category: neutral currents, the behavioural effects of frontal lesions, and the melting point of lead are a heterogeneous lot, and they are explicitly agnostic about whether there is a unified metaphysics of phenomena (§III). The distinction they want is *functional* rather than ontological: phenomena are stable, recurrent features of the world that are appropriate targets of systematic explanation; data are tied to particular apparatus and particular occasions and bear the imprint of many idiosyncratic factors that no general theory aims to address (§IV). The functional distinction can do the relevant work without a sharp categorical line — and indeed without any commitment about the ultimate ontological status of phenomena.

O4 *Over-generalizations: “phenomena are typically not observable,” “theories never explain data,” ... Are these really required for the argument?* (Woodward 2011, § 3)

Woodward concedes: these formulations were over-stated and turn out to be unnecessary. There are plausible cases of observable phenomena (Paul Teller's examples: colour constancy, pop-out attention effects, the Poisson spot), and there is no need to deny that theories ever explain data. What the framework requires, and all it requires, is that data-to-phenomena reasoning *can* (and not infrequently does) succeed without explanatory derivation of the data. The residual claim is structural and pluralist: different scientific contexts call for different mixes of theory-driven and data-driven inference, and the original distinction earns its keep by giving us a vocabulary in which to describe the data-driven cases.

O5 *Share the language of “phenomena” with Duhem — B & W's title deliberately echoes this — but how do their views compare?*

Duhem's “phenomena” are roughly the experimental laws that physical theories aim to “save” through their *representational* (classificatory, mathematical) part. B & W's phenomena are stable features of

the world distinguished from the data that constitute evidence for them. Duhem’s “phenomena” are closer to what B & W call data. Duhem’s contrast between the representational and the explanatory parts of a theory orthogonal to B & W’s contrast: Duhem’s representational part of theory *does* the explaining, like what B & W mean by systematic explanation of phenomena. Both Duhem and B & W reject naive views about how theories relate to phenomena — but they reject different naive views and offer different replacements. Which framework is better, and more compatible with the fine-grained evidential reasoning Smith identifies in historical case studies?

O6 *Comparison to Smith and Smith-Seth’s analysis of evidential reasoning?*

For B & W, reliability assessment is local: confounder control, calibration, statistical analysis of apparatus and runs of measurements; Smith’s long term view of evidence seems to run counter to this. Smith & Seth distinguish “taking a hypothesis to be true” (a working presumption supporting further inquiry) from belief licensed by inference to the best explanation. Where in B & W’s picture, if anywhere, does this distinction find a home? Is Woodward (2011)’s § 6 concession enough to accommodate theory-mediated measurement?

Looking Ahead: Issues for Woodward (2011)

Woodward’s restatement clarifies the framework, walks back several over-statements in the original, and addresses critics (Glymour, McAllister, Schindler, Votsis). Some questions worth keeping in mind:

- *Corrections to 1988.* Woodward concedes that some claims (“phenomena are typically not observable”; “theories don’t explain data”) were over-stated and unnecessary. How does softening these claims impact the overall thesis?
- *The role of background assumptions.* Inductive inference always has the form *evidence + substantive empirical assumptions* → *conclusion*. Data-to-phenomena reasoning is no exception. Are B & W really vulnerable to a “radical inductivist” reading (Schindler)? What is the difference between the additional assumptions involved in data-to-phenomena reasoning and the systematic explanatory theories whose role is being denied?
- *McAllister’s challenge.* If additional assumptions are always required, do we end up with a “radically polymorphic” world, in which any data set realizes indefinitely many phenomena depending on the investigator’s choice of noise level? Woodward distinguishes the role of *interests / attitudes toward risk* from that of substantive empirical assumptions about, e.g., error distributions. Is this distinction stable?
- *Theory-involvement without circularity.* The Eddington and Millikan cases: theory plays a motivating role and supplies vocabulary, without thereby explaining individual data. Woodward’s schematic case (theory *T* predicts phenomenon *P*, data *D* at the limits of available technique): *T*’s prediction can legitimately raise our confidence that error has been controlled. When is such “mutual reinforcement” viciously circular, and when is it not?

References

Hempel, C. G. (1952). Fundamentals of concept formation in empirical science. In Neurath, O., Carnap, R., and Moris, C., editors, *Foundations of the unity of science, Volume II* No. 7, pages 1–52. University of Chicago Press, Chicago.